

SPINE

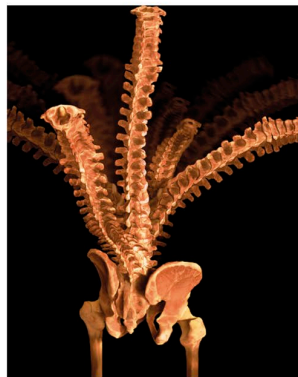
THE SPINE IS ALSO KNOWN AS THE SPINAL OR VERTEBRAL COLUMN, OR SIMPLY "THE BACKBONE". THIS STRONG YET FLEXIBLE CENTRAL SUPPORT HOLDS THE HEAD AND TORSO UPRIGHT, YET ALLOWS THE NECK AND BACK TO BEND AND TWIST.

SPINE FUNCTION

The spine consists of 33 ring-like bones called vertebrae. The bottom nine vertebrae are fused into two larger bones termed the sacrum and the coccyx. The 26 movable components of the spine are linked by a series of mobile joints. Between the bones of each joint is an intervertebral disc – a springy pad of tough, fibrous cartilage that squashes slightly under pressure to absorb shocks. Ligaments and muscles around the spine provide stability and help to control movement. The spinal column also protects the spinal cord (see p.98).

FLEXIBLE COLUMN

Owing to the shape of the vertebrae, the spine can bend further forward than back, and twist on its axis.



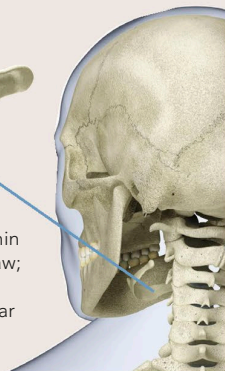
HYOID BONE

The single U-shaped hyoid bone is located at the root of the tongue, just above the larynx. It is one of the few bones in the body that does not join directly to another bone. It is held in position by muscles and by the strong stylohyoid ligament on each side of the bone, which links to the styloid process of the skull's temporal bone. The hyoid stabilizes several sets of muscles used in swallowing and speech.



LOCATION

The hyoid bone sits within the curve of the lower jaw; it has two pairs of small horn-like projections near the front.

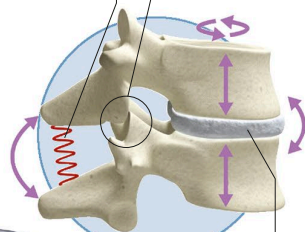


Springy ligament

Ligaments between spinous processes limit movement and store energy for recoil

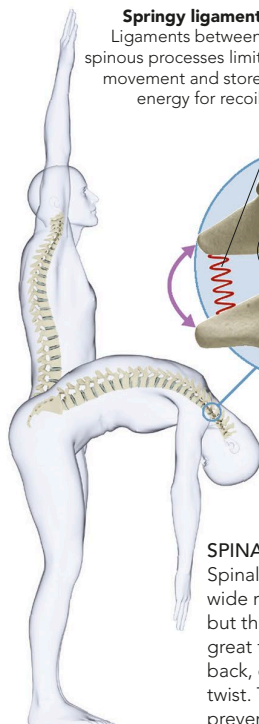
Facet joint

Determines range of movement between vertebrae



Intervertebral disc

Composed of tough, flexible fibrocartilage with jelly-like core



SPINAL JOINTS

Spinal joints do not have a wide range of movement, but they still allow the spine great flexibility, letting it arch back, curve forward, and twist. Two facet joints help to prevent slippage and torsion.